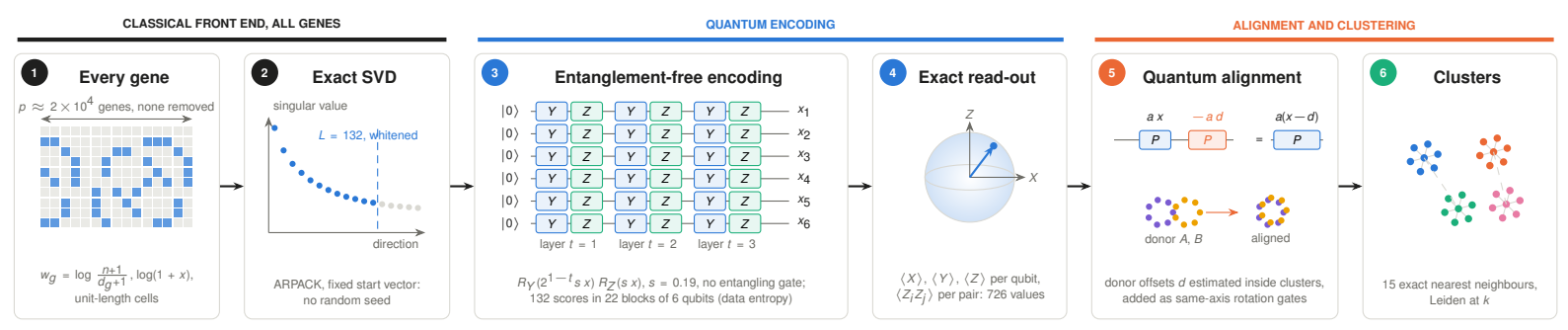
CLASSICAL FRONT END, ALL GENES
QUANTUM ENCODING
ALIGNMENT AND CLUSTERING
1
Every gene
$p \approx 2 \times 10^4$ genes, none removed
$w_g = \log \frac{n+1}{d_g+1}$, $\log(1+x)$,
unit-length cells
2
Exact SVD
singular value
L = 132, whitened
direction
ARPACK, fixed start vector:
no random seed
3
Entanglement-free encoding
$|0\rangle$ Y Z Y Z Y Z x_1
$|0\rangle$ Y Z Y Z Y Z x_2
$|0\rangle$ Y Z Y Z Y Z x_3
$|0\rangle$ Y Z Y Z Y Z x_4
$|0\rangle$ Y Z Y Z Y Z x_5
$|0\rangle$ Y Z Y Z Y Z x_6
layer t = 1
layer t = 2
layer t = 3
$R_Y(2^{1-t}\, s\, x)$ $R_Z(s\, x)$, s = 0.19, no entangling gate;
132 scores in 22 blocks of 6 qubits (data entropy)
4
Exact read-out
Z
X
$\langle X \rangle$, $\langle Y \rangle$, $\langle Z \rangle$ per qubit,
$\langle Z_i Z_j \rangle$ per pair: 726 values
5
Quantum alignment
a x
− a d
a(x − d)
P
P
=
P
donor A, B
aligned
donor offsets d estimated inside clusters,
added as same-axis rotation gates
6
Clusters
15 exact nearest neighbours,
Leiden at k